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Breast Cancer Prediction using Optimized Machine Learning Classifiers and Data Balancing Techniques

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***Abstract*— The most prevalent malignancy in females and the second greatest reason for the loss of life from cancer is Breast Cancer. Hence, a computer assisted detection (CAD) system that uses a machine learning technique to give reliable breast cancer diagnosis is required. The paper is aimed to incorporate several machine learning (ML) algorithms, including Support Vector Machine (SVM), Logistic Regression, k-Nearest Neighbour (KNN), Decision Tree (DT), Random Forest (RF), Artificial Neural Network (ANN), and Nave Bayes (NB) with hyperparameter tuning using the Random Forest Feature Importance Method for feature selection. These models have been trained on the Wisconsin Diagnostic Breast Cancer (WDBC) Dataset. Furthermore, the dataset was balanced using both Undersampling and SMOTE from which we concluded that Undersampling gave us an overall better result. The performance evaluation parameters for the designed model are specificity, accuracy, sensitivity, F1 score, precision, recall and Area Under the Curve(AUC). The research concluded that the Support Vector Machine Algorithm proved to be the most effective model which fit our dataset with an Accuracy of 95.8% followed by KNN with an accuracy of 95.3%.**

Keywords—Breast Cancer, Machine Learning, Wisconsin Breast Cancer Data Set, Undersampling, Random Forest Feature Selection

I. INTRODUCTION

Today, the second greatest reason for the loss of life in all countries worldwide is Cancer and is a major obstacle to life expectancy. Early-stage cancers are easier to treat and patients have a higher chance of survival as compared to later- stage cancers. The most common malignant disease in women, often characterized by a lack of early symptoms is breast cancer, resulting in delayed detection of the disease. Research shows that at an advanced stage of the disease, the treatment becomes more difficult and uncertain.

Breast cancer among Indian women accounts for 25-32% of all cancer cases for Indian urban female population (according to urban population-based cancer registries) and the second most populous cancer in rural areas. In cities such as Mumbai, Bhopal, Delhi, Chennai, Kolkata, Ahmedabad and Bengaluru, breast cancer contributes to more than 1/4th of all female cancer cases i.e. about 25% to 32% of all female cancers. The 25-50 years age group accounted for about 50% of all cases were, and for the advanced cases more than 70% had high mortality and low survival. The number is overwhelming and is increasing constantly. Breast cancer can be treated, and if detected early, it is more likely to survive. The only way to do this is by the diagnosing it in the early stages, followed by an early prognosis of this cancer type as it eases patient's succeeding clinical administration. As per globocon 2018 [23] cancer statistics for Indian female, majority of the cases accounts for breast cancer which is shown in fig 1.

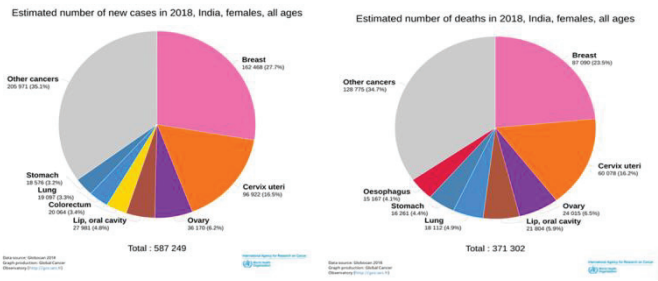

Fig. 1. Cancer statistics among women in India as per Globocan 2018 [25]

Over the last few decades, continuous evolution has been made in connection with cancer research, especially in breast cancer. Scientists have made use of a variety of methods, including early screening to find cancer before it's symptoms visible. Accompanied with the introduction of novel technologies in the field of medicine, a lot of cancer statistics has not only been acquired but also been made accessible to the medical community of research. Machine Learning methods have as a result become a prominent device for researchers in the medical field as they help to identify, discover relationships and patterns from complex datasets and effectively make future predictions of cancer types.

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This research utilizes the Wisconsin Breast Cancer Dataset (WBCD) and various ML classification techniques are applied in order to find out the form of breast cancer in a patient. Eight classification models comprising of, namely Logistic Regression, k-Nearest Neighbour (KNN), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), Artificial Neural Network (ANN) and, Naïve Bayes (NB) were utilized. The prime objectives of the research are:

- Analysing and balancing WBCD for detecting feature relations
- Selecting only the most important features using ML techniques.
- Applying various established classification techniques to the balanced dataset to classify into benign and malignant.
- Discovering the best suited algorithm with the highest prediction accuracy after comparing results.

II. Literature Survey

Recently, several studies were performed to detect and classify Breast Cancer using ML models to improve accuracy. Logistic Regression, SVM, KNN, RF, DT, NB, Artificial Neural Network (ANN), XGBoost (XG), Extreme Learning Machine (ELM) and Ada-Boost (AB) are examples of machine learning algorithms implemented by several researchers. These studies have been conducted using a variety of datasets, including the WBCD from the UCI ML Repository, Digital Database for Screening Mammography (DDSM), Breast Cancer Coimbra dataset (UCI) and BreakHis dataset. To complete their research, authors extract and select various characteristics from these datasets. In the study conducted by David Hettich et al. [6], one thousand six hundred and sixty four breast cancer tumor pictures that are pre-classified from the DDSM have been explored. Researchers have selected 4 distinctive ML strategies specifically SVM, ANN, KNN and RF. Their findings showed that SVM and ANN have shown to have 93%, the highest accuracy-score.

Shital Pawar [17] worked on six different classification models. They have compared SVM, KNN, DT, RF, XG and AB to find out which method performs the best in terms of accuracy. XB proved to give the highest accuracy 98.24%. Therefore, XB had the best performance.

Telsang and Hegde [11] chose five methods of Machine Learning to measure their effectiveness in predicting the type of tumor. These models were: NB, RF, SVM, LR and KNN amongst them SVM proved to have the highest accuracy of 96.25.

Sharma and Mehra [18] Compared performance criteria for supervised learning classifiers RF, SVM, LDA, transfer learning techniques (VGG16, VGG19, ResNet50 as a feature extractors); Find the best classification elements in the BreakHis Breast Cancer dataset. Experiments have shown that VGG16 + SVM (L, 1) bears the highest results of all magnifications (x40, x100, x200, x400).

The author P. Gupta and S. Garg used KNN, LR, DT, RF, SVM and ANN and achieved 98.9% as the highest accuracy for the Deep Learning technique ANN [20]. In the paper, Standard ELM algorithm was optimized for the highest accuracy rate of 80% and the lowest training period. Sharma et al. [13] used RF, NB, KNN and DT algorithms. Each of the algorithms had an accuracy of more than 94% with KNN being the most effective in detecting breast cancer since it had the highest accuracy of 95.90% with a precision of 98.27% and F1 score of 94.20% over the other algorithms. Khourdifi and Bahaj [22] proposed a solution by comparing various algorithms such as NB, RF, SVM and KNN for the diagnosis with the highest accuracy shown by SVM of 97.9%. The benign category received the largest TP of 99.7%, whereas malignant category received 94.6%. When it involves classifying breast cancer data, it outshines other with respect to accuracy, precision, sensitivity and specificity.

Machine learning algorithms struggle to extract reliable information from data that is skewed. The study of unbalanced datasets can lead to biased results. Feature selection is important in reducing the amount of variables input and examine only the most significant variables. The selection of features is important to decrease the number of input variables and consider only the most significant ones. From the papers that we reviewed for our research, none of them had implemented feature selection or any of the balancing dataset techniques on the WBCD. Thus, the primary focus of the research is comparing the various model results after performing Random Forest Feature Importance Method for feature selection on a balanced dataset using under sampling.

III. Proposed Methodology

The methodology suggested is given in fig shown below.

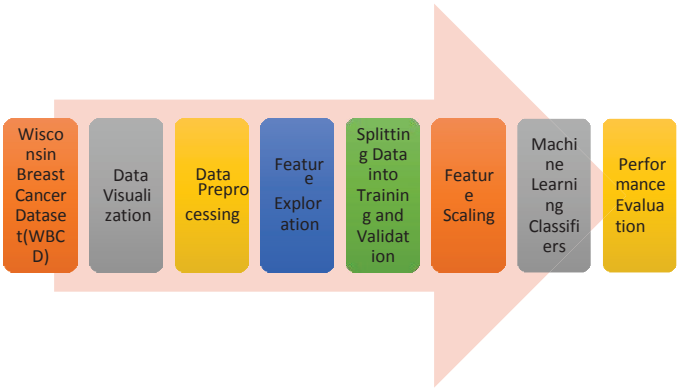

Fig. 2. Workflow of Proposed Methodology

For the purpose of this research, the WBCD has been imported followed by performing data visualization to get a better understanding of the dataset. The next step was data preprocessing which included the label encoding, balancing the dataset and feature extraction. The categorical variable 'Diagnosis' was converted into numerical datatype with Malignant being 1 and Benign being 0. For balancing the dataset, both undersampling and SMOTE were implemented and it was observed that Undersampling gave better results. The Random Forest Feature Importance Method and was used and the 7 most important features that had a high correlation with the dependent variable 'Diagnosis' were selected. The dataset was split into training (75%) and validation (25%) before feature scaling. The StandardScaler method from SciKit-Learn library was used for feature

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scaling. Finally, different models were trained on this dataset and their respective results were compared.

IV. DATASET USED

The research utilized the WBCD. The 10 real-valued attributes used to describe every instance are:

a) radius: mean of distances from centre to points on the perimeter,
b) texture: standard deviation of greyscale values,
c) perimeter,
d) area,
e) smoothness: local variation in radius lengths,
f) compactness= perimeter^2 /(area 1.0),
g) concavity: severity of concave portions of the contour ,
h) concave points: number of concave portions of the contour,
i) symmetry,
j) fractal dimension= (“coastline approximation" 1)

Fig. 3. Real-valued features

The instance ID attribute was removed during the feature extraction. In addition, there is one class attribute which consists of two classes- Malignant (M) and Benign (B). Additional data analysis revealed thirty features with five hundred and sixty nine occurrences, three hundred and fifty seven of which belong to the benign category and two hundred and twelve of which belong to the malignant category.

A. Exploratory data Analysis

The data was highly imbalanced with 568 data entries divided into 2 classes (Malignant- 211, Benign- 357). No null or missing values were found. To balance the dataset, the Undersampling and SMOTE techniques were implemented.

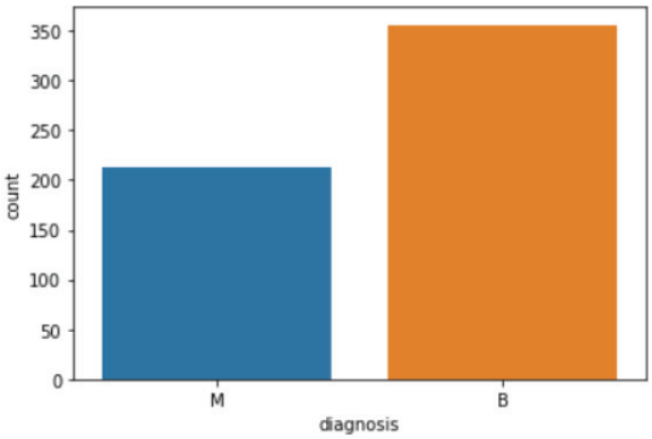

Fig. 4. Class imbalance in WBCD dataset

B. Dataset Balancing Techniques

Undersampling: It is a technique for equalizing non-uniform data sets by keeping all data in minority bins and reducing the size of the majority bins. To put it another way, the technique removes samples from training datasets that belong to the dominant class to improve the class distribution'sbalance.

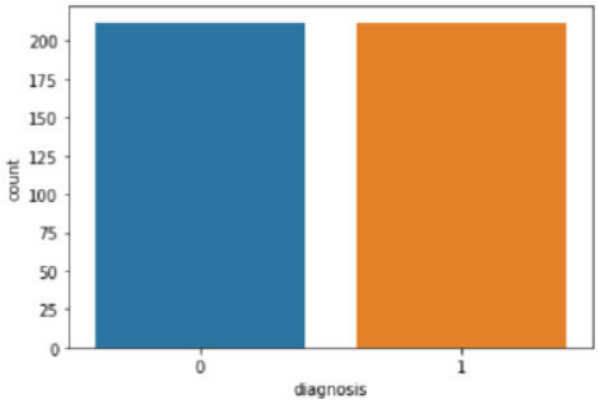

Fig. 5. Class balance using undersampling data balancing technique

SMOTE (Synthetic Minority Oversampling Technique): SMOTE functions by choosing a close-to-the-feature-space example, drawing lines in between attribute-space examples, and finding patterns. Particularly, an arbitrary sample of the minority class is initially generated. The k-nearest neighbour (typically K = 5) is then found in this example. A random neighbour is selected, a combined instance is generated at a random location in the attribute space between the two examples. The dataset was distributed as follows after executing SMOTE:

```
0    357
1    357
Name: diagnosis, dtype: int64
```

Fig. 6. Class balance using SMOTE data balancing technique

Of the two methods that are mentioned above, it was observed that balancing the dataset using Undersampling gave better results.

C. Feature Selection

In this work, feature selection is done using statistical analysis and using random forest feature importance method.

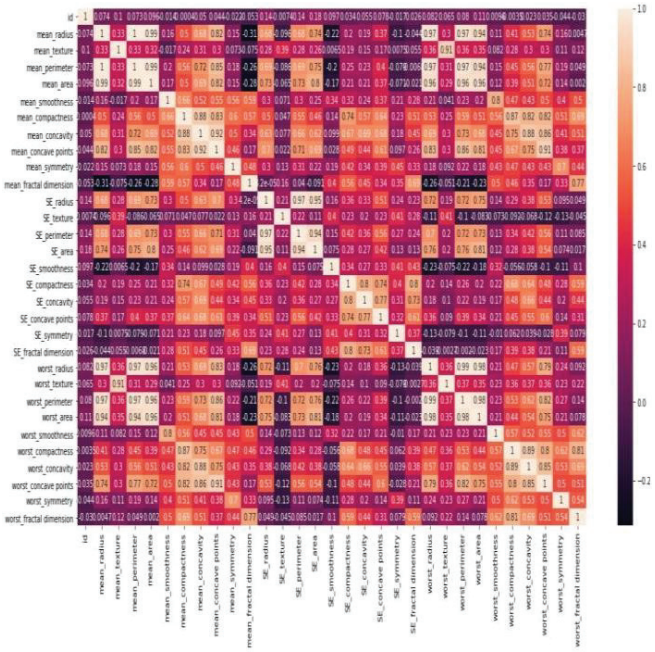

Fig. 7. Heatmap for feature Selection using statistical correlation

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The heatmap failed to give a clear relationship between the dependent and independent variables in a straightforward way. As a result, the Random Forest Feature Importance Method was put into practise. The horizontal bar plot was created after completing the Random Forest Feature Importance Method, as shown in fig 8, and the most contributing features were picked based on that.

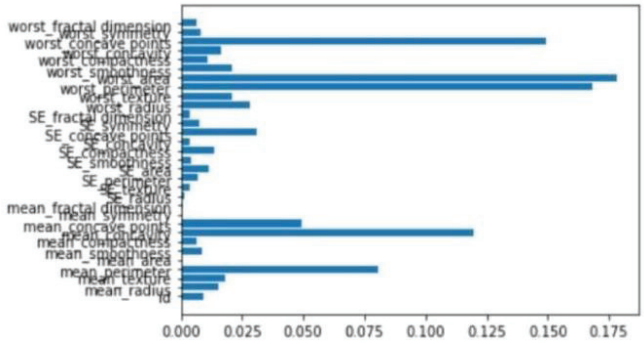

Fig. 8. Feature Selection using Random Forest Importance method

The eight most important features selected were SE_concave points, mean_concavity, worst_area, mean_perimeter, mean_concave points, worst_concave points, worst_radius and worst_perimeter.

V. RESULTS AND DISCUSSION

The machine learning model performance is evaluated based on different evaluation parameters which include the followings:

Accuracy: Accuracy is understood as the number of correctly predicted for a complete test dataset that the model has made. Accuracy is a basic good indicator for measuring the model performance. To calculate accuracy, use the following formula:

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

Where TP =True Positive, TN = True Negative, FP = False Positive, FN = False Negative

Area Under the Curve (AUC): AUC is defined as the ROC curve summary measuring the ability of a classifier to detect different classes. It shows how precisely the model can classify negative and positive category. The model with greater AUC is better.

$$AUC = \frac{SE + SP}{2}$$

Where SE = Sensitivity, SP =Specificity

F1-score: It is a combination of both recall and precision providing one measure. Mathematically it is the harmonic mean of precision and recall that can be calculated as follows:

$$F1\,Score = (2 * Recall\ *\ Precision)\ /\ (Recall + Precision)$$

Sensitivity: True positive rate, is a measure of the proportion of people with an illness that is correctly predicted to have the illness. Sensitivity can also be expressed in the form of true positive rates (TPRs). The higher the true positive rate, the better the model is in correctly identifying positive cases. To calculate specificity, use the following formula:

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity: Specificity is defined as the percentage of actual negatives predicted to be negative (or true negative) that shows there exists another percentage of true negative results that can be predicted as positive and labelled as false positives, also called false positive rate. Formula for calculating specificity is:

$$Specificity = \frac{TN}{TP + FP}$$

Optimizable Machine learning Classifiers

Machine learning hyperparameter optimization or optimizable machine learning classifiers aims to find hyperparameters for a specific machine learning algorithm that provides the best performance when measured against a validation set. It helps maximize the predictive accuracy of the model and minimize the cost function.

Classifiers	Tuned Hyperparameters
LR	-
SVM	Kernel function: Quadratic Box constraint level: 29.3215 Multiclass method: One-vs-One Standardize data: True
KNN	Number of neighbors: 3 Distance metric: City block Distance weight: Inverse Standardize data: true
RF	Maximum number of splits: 423 Number of learners: 30
DT	Maximum number of splits: 4 Split criterion: Gini's diversity index
NB	Distribution names: Kernel Kernel type: Gaussian
ANN	Number of fully connected layers: 3 First layer size: 10 Second layer size: 10 Third layer size: 10 Activation: ReLU Iteration limit: 1000 Regularization strength (Lambda): 0 Standardize data: Yes

Fig. 9. Optimizable Machine Learning Classifiers with best set of hyperparameters used in proposed work

The confusion matrix and ROC curve for each classifier is shown in Fig 10.

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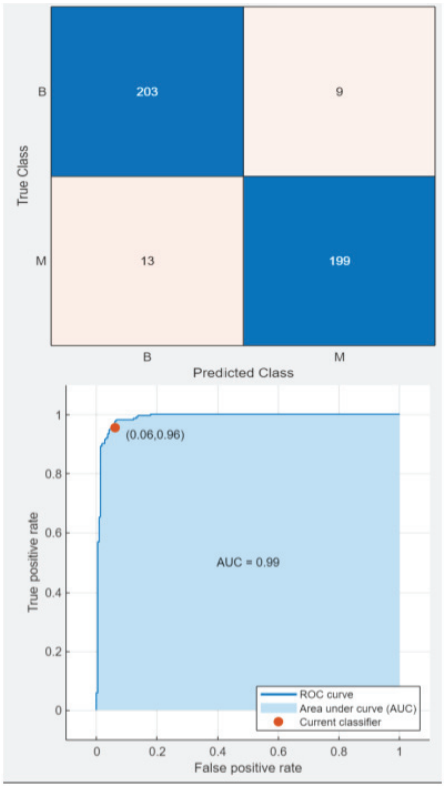
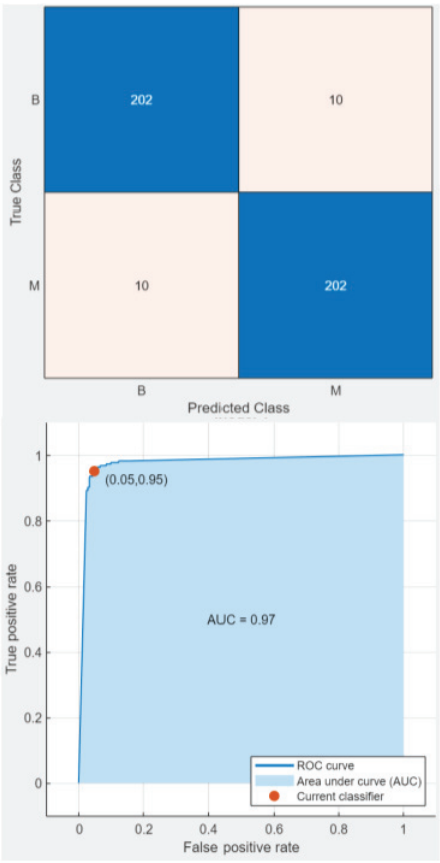
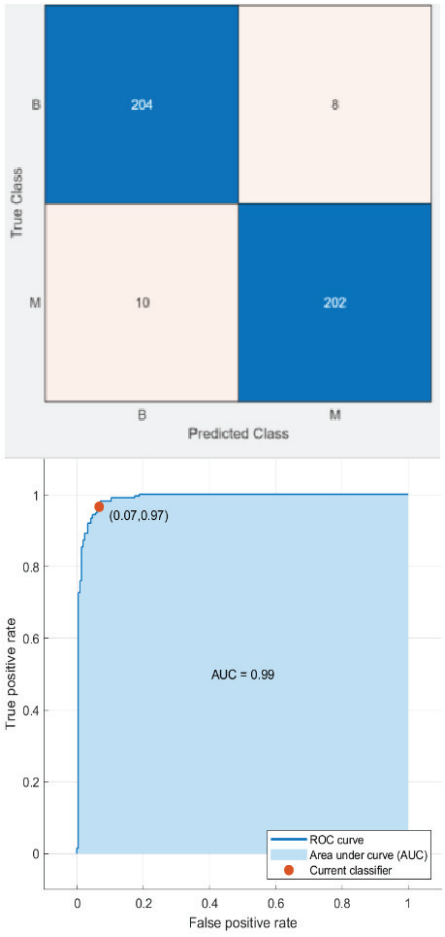


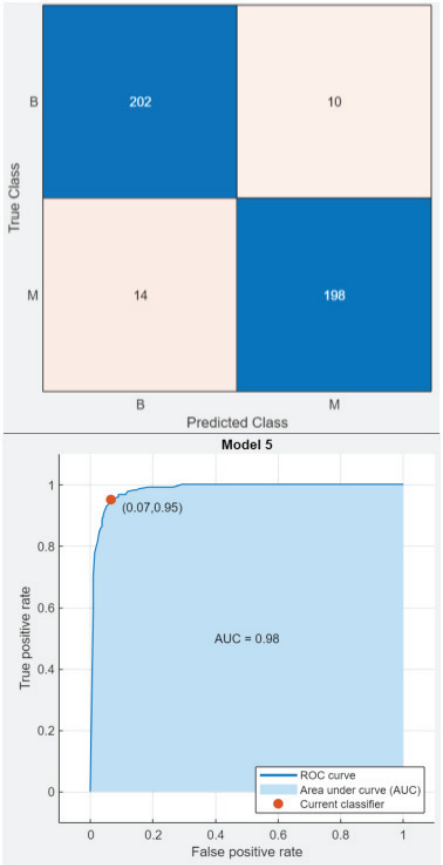
(a) LR



(c) KNN



(b) SVM



(d) RF

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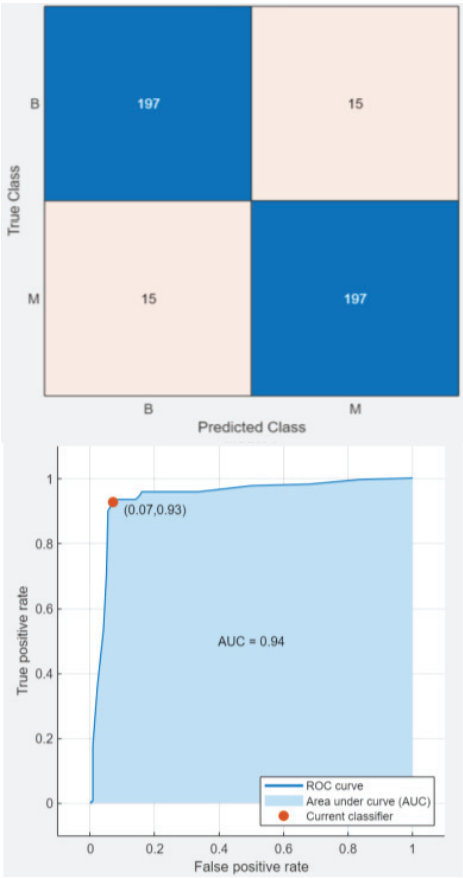
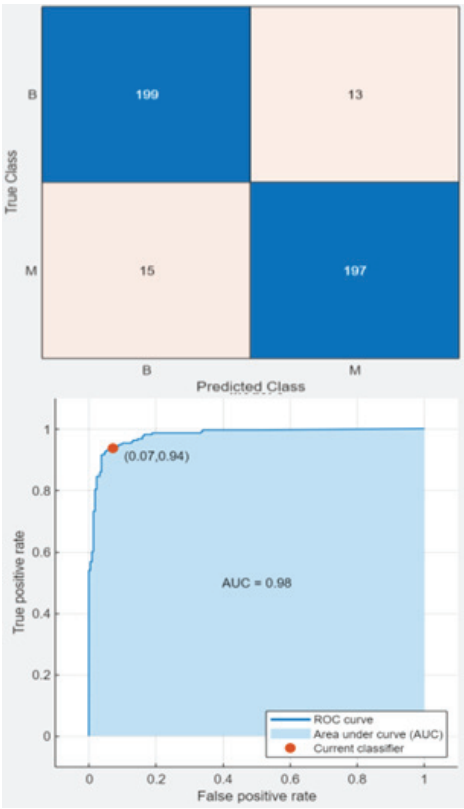
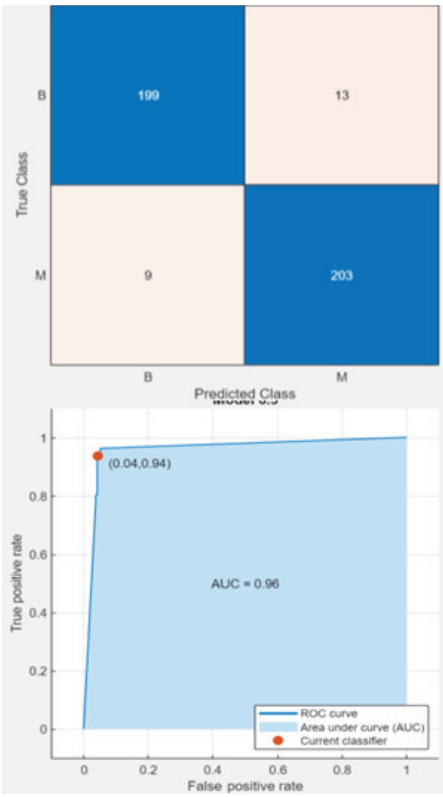


(e) DT



(f) NB



(g) ANN

Fig. 10. Confusion matrix and ROC curve for all classifiers

The models comparison is displayed in Table 1 and Fig 11 presents a bar chart comparing accuracies for all the seven machine learning algorithms. Among all the algorithms, SVM using quadratic kernel gave the best results with accuracy of 95.8% followed by KNN with an accuracy of 95.30. Both show a high value for area under the ROC curve i.e., 0.99, 0.97 respectively.

TABLE I. COMPARATIVE RESULT ANALYSIS

ML Model	Accuracy	ROC	F1-Score	Sensitivity	Specificity
LR	95%	99%	94.86%	93.98%	95.67%
SVM	95.8%	99%	95.77%	95.33%	96.19%
KNN	95.3%	97%	95.28%	95.28%	95.28%
RF	94.3%	98%	94.39%	93.52%	95.19%
DT	92.9%	94%	92.92%	92.92%	92.92%
NB	93.4%	98%	93.43%	92.99%	93.81%
ANN	94.3%	96%	94.76%	95.67%	93.98%

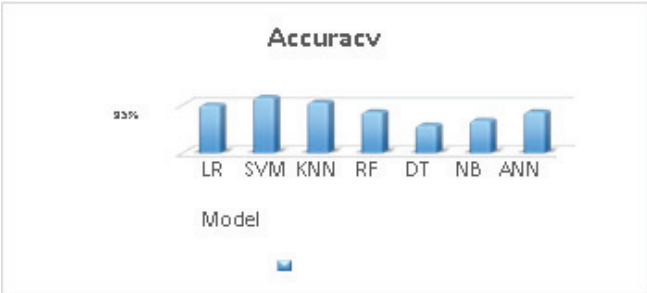


Fig. 11. Accuracy For Different Models

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VI. CONCLUSION AND FUTURE WORK

The usage of seven distinct machine learning approaches was proposed in this study to create a balanced breast cancer diagnosis model, also we extracted seven features that use the Random Forest Feature Importance Method for feature selection. The models were compared based on their accuracies. Subsequently, all other measures like sensitivity, specificity, F1-score and ROC curve were also studied. Since the goal of our project is to identify everyone who has breast cancer, the number of false negatives should be low and hence we have a high sensitivity. This is especially important in our case since the consequence of failing to treat the condition are serious and very effective. Depending on the hyper parameters chosen, each algorithm works differently. In terms of overall methodology, SVM came up on top, followed by the KNN technique.

In the future, various deep learning algorithms can be implemented to detect not only the different stages but also the different categories of breast cancer thereby enhancing the existing system's performance. Another future research prospect could be to apply the above ML algorithms in E-health.

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